3D Roof Reconstruction with Computer Vision for Solar Energy Optimization

Date:

Table of Contents

1. Executive Summary

IECO is developing a Proof of Concept (PoC) for an AI-driven system that automates the 3D reconstruction of roofs from aerial images and digital surface models (DSMs). The goal is to streamline the design and simulation of photovoltaic (PV) installations, which currently requires users to manually trace roof perimeters and input volumetric data. This manual process is not only time-consuming but also vulnerable to human error, impacting both efficiency and accuracy.

The solution intends to leverage AI and point cloud data from DSMs to automatically generate precise 3D models of industrial and commercial roofs. This capability will allow users to create accurate simulations and engineering analyses with greater ease and speed. In turn, this will support the broader objectives of advancing photovoltaic self-consumption and enabling the growth of energy communities. By reducing the time and cost associated with the solar project design workflow, the solution will contribute significantly to accelerating sustainable energy transitions.

With a focus on accuracy, efficiency, and scalability, this PoC aims to transform the workflow for solar PV projects, enabling users to model roofs faster and more accurately, reducing error rates, and ultimately driving the adoption of solar energy technologies more effectively. This automation represents a key step toward making solar energy deployment faster, scalable, and more accessible, advancing the global shift toward renewable energy.

2. Introduction & Problem Statement

The adoption of photovoltaic (PV) installations plays a critical role in achieving a fair and sustainable energy transition by reducing dependence on fossil fuels and promoting renewable energy solutions. As global efforts intensify to combat climate change and transition to greener energy systems, the demand for efficient PV system design and deployment continues to grow. A key component of PV system design involves creating 3D roof models to assess solar energy potential, optimise installations, and ensure cost-effectiveness. However, traditional methods for generating these models rely on manual processes, which are slow, resource-intensive, and error-prone. To address these limitations, this project proposes an AI-driven solution to automate 3D roof modelling, providing a faster, more reliable, and scalable approach to accelerating PV adoption and advancing global sustainability goals.

Currently, the manual creation of 3D roof models for PV design presents significant challenges. This process is time-consuming and labour-intensive, requiring substantial human effort and expertise. Moreover, it is prone to errors, which can negatively impact project accuracy and efficiency, leading to costly delays in PV system deployment. These inefficiencies hinder the development workflow, increasing costs and slowing down the widespread adoption of solar energy systems. By implementing an AI-driven solution, this project aims to automate 3D roof modelling, enabling faster and more accurate roof reconstructions while reducing costs and improving the overall user experience. This automation will streamline PV design and simulation, accelerating the deployment of renewable energy systems and supporting the global transition toward a more sustainable future.

3. Objectives

Automate 3D Roof Modelling

- ➢ Develop an AI-driven solution to replace manual 3D roof modeling processes, ensuring faster and more efficient roof reconstructions.

Enhance Modelling Accuracy

- ➢ Improve the precision of 3D roof models by leveraging advanced AI techniques, minimising errors and inconsistencies in design outputs.

Streamline PV System Design Workflow

- ➢ Reduce the time and effort required to create 3D roof models, enabling seamless integration into PV design and simulation processes.

Reduce Costs and Increase Accessibility

- ➢ Lower the costs associated with manual modelling and make renewable energy deployment more accessible to a broader range of users.

Support Renewable Energy Transition

- ➢ Accelerate the adoption of PV installations by providing reliable, scalable, and cost-effective tools, contributing to global sustainability and energy transition goals.

Deliver Measurable Outcomes

- ➢ Ensure measurable improvements in modelling time, accuracy, and cost-efficiency to evaluate the project's success and impact on PV adoption.

4. Project Methodology

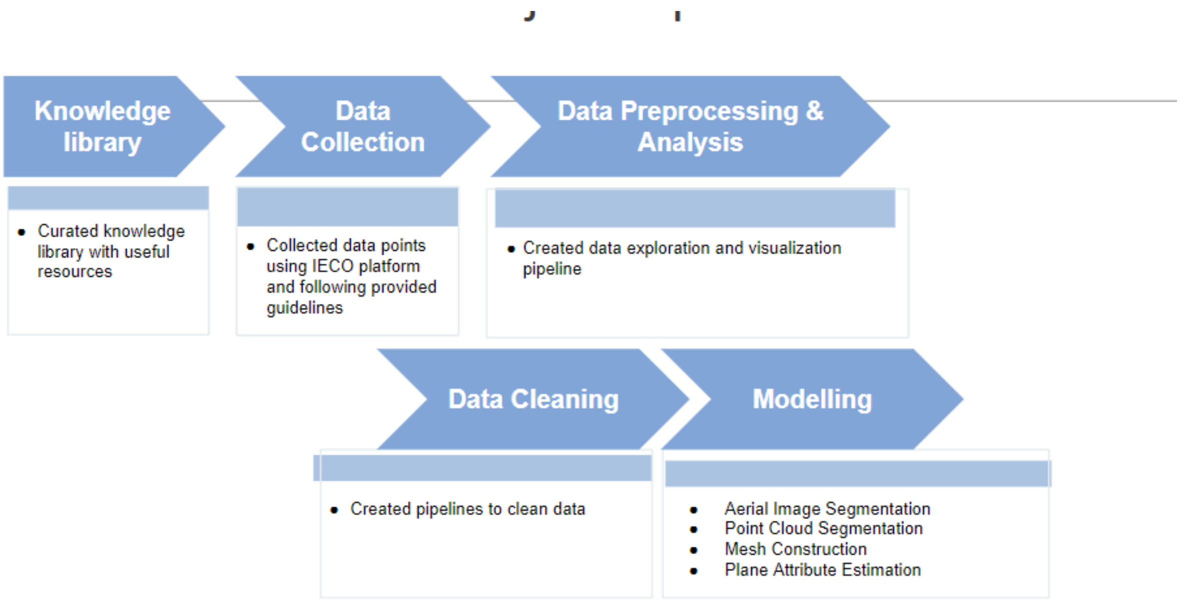

Figure 1: Project pipeline

We followed a structured pipeline for this project, starting with the development of a **Knowledge Library** to provide a curated repository of resources. A curated **knowledge library** was created, containing useful resources and references. This ensured that all team members had access to reliable materials throughout the project.

Here is a link to the knowledge library sheet.
https://docs.google.com/spreadsheets/d/19hJaNrpyOmrfOSXGIRhg2X8mMy__KBnuervQBTKG2Pc/edit?gid=124186218#gid=124186218

Next, the **Data Collection** phase involved gathering aerial imagery and corresponding point cloud data using the IECO platform, adhering to the guidelines provided by the partner to ensure data relevance and quality.

In the **Data Preprocessing & Analysis**, a data exploration and visualisation pipeline was created to assess the data, identify inconsistencies, and gain insights into its structure.

This was followed by the **Data Cleaning** phase, where custom pipelines were developed to remove noise, correct errors, and standardise the data, ensuring it was ready for modelling.

The final stage, **Modeling**, employed a variety of techniques, including aerial image segmentation, point cloud segmentation, mesh construction, and plane attribute estimation. These models enabled 3D roof reconstructions, streamlining workflows and providing essential insights into roof structures.

4.1 Data Collection & Preprocessing (Rosho)

[Describe how the data was collected and prepared for analysis. Include any challenges faced during this stage and how they were mitigated.

Instructions: Provide a detailed explanation of the data sources, the preprocessing steps taken, and any tools used. Highlight any issues encountered and how they were resolved.]

Insert any tables or figures representing the data collection process or preprocessing steps here.

4.2 Modelling (Louis)

[Explain the techniques used to analyse the data or build models. Include any algorithms or tools utilised and justify their selection.

Instructions: Detail the algorithms or models used, including the rationale behind their selection. Include performance metrics and other relevant details.]

Insert any relevant tables or figures depicting model performance or analysis results here.

Proposed Solution

The proposed solution involves two different approaches. The first approach begins with Image Masking, where roof polygons are used as masks on aerial images. This approach highlights roof areas, enabling the model to focus exclusively on features within these regions. Once roof areas are identified, Point Cloud Extraction maps the segmented 2D regions onto 3D point cloud data, using spatial alignment such as GPS coordinates. This step ensures that only the points corresponding to roof areas are extracted for further processing. Finally, the 3D Reconstruction process uses the spatial points from the point cloud to generate accurate 3D roof models.

The second modelling approach focuses on **Plane Attribute Estimation**, leveraging advanced convolutional neural networks (CNNs) like EfficientNet or ResNet50 to predict key roof parameters, including tilt, azimuth, and height. This method combines features extracted from aerial images and point cloud data to deliver accurate attribute predictions. The process begins with standard preprocessing and data augmentation, enhancing the robustness of the model by introducing variations such as scaling and rotation. A CNN backbone is then utilised to extract high-level features from the aerial images, which are concatenated with the corresponding point cloud data to integrate spatial and visual information. Custom loss weights are incorporated into the training process to prioritise the accurate prediction of specific attributes, ensuring the model is tailored to the unique requirements of roof parameter estimation. This approach enables efficient analysis of roof planes, facilitating downstream tasks such as PV system design and 3D modelling.

Modelling Experiments

4.3 Deployment Strategy

[If applicable, describe how the project's findings or models were deployed in a real-world setting.

Instructions: Outline the deployment process, including any platforms or tools used. Discuss any challenges encountered during deployment.]

5. Results & Key Findings

[Present the project's findings, including both quantitative and qualitative results.

Instructions: Provide a detailed account of the project's outcomes, using tables and figures to present data where applicable. Include a narrative that explains the significance of the results.]

Insert tables, graphs, or other visual representations of the results here.

6. Challenges & Lessons Learned

[Discuss any challenges faced during the project and the lessons learned.

Instructions: Highlight key challenges and how they were addressed. Discuss any insights gained that could be valuable for future projects.]

7. Conclusion & Recommendations

[Summarise the project's overall findings and provide recommendations for future work or potential applications of the results.

Instructions: Conclude the report by summarising the main findings and providing actionable recommendations based on the project's outcomes.]

8. Future Work

[Suggest areas for future research or development based on the project's findings.

Instructions: Propose further studies, experiments, or iterations that could build on the work done in this project.]

9. Appendices *(Optional)*

[Include any supplementary material that supports the report, such as detailed data tables, code snippets, or additional references.

Instructions: Appendices should contain material that is referenced in the report but is too detailed to include in the main sections.]

Instructions for Filling the Report:

- **Tables & Figures:** Ensure all tables and figures are numbered and include descriptive captions. Mention these in the text before they appear.
- **Data Sources:** Reference all data sources accurately.
- **Formatting:** Use consistent formatting throughout the report. Ensure headings, subheadings, and body text are clear and easy to navigate.
- **Clarity:** Write clearly and concisely, avoiding jargon unless necessary. Define any technical terms or acronyms whe hbn first introduced.